

Dataset 3: Multiple Linear Regression

Student Academic Performance Prediction

1. Research Objective

This study models final exam performance of 150 students as a linear function of five academic and lifestyle predictors. The objective is to identify which factors most strongly predict exam success and quantify their independent contributions while controlling for all other variables. Sleep hours are modeled with a quadratic term to capture the inverted-U relationship between sleep and cognitive performance.

2. Variables

Role	Variable	Description
Dependent	Final_Exam_Score	Final exam score (continuous, 0–100)
Independent	Study_Hours	Average study hours per week
Independent	Attendance_Rate	Class attendance percentage
Independent	Sleep_Hours	Average sleep hours per night (linear + quadratic)
Independent	Stress_Level	Self-reported stress score
Independent	Practice_Tests	Number of practice tests completed

3. Methodology — Multiple Linear Regression

Multiple linear regression is appropriate because the dependent variable (exam score) is continuous and we aim to quantify the effect of each predictor while holding others constant. The quadratic Sleep_Hours term is included based on documented non-linear cognitive-sleep relationships. Standardized beta coefficients allow direct comparison of predictor importance. VIF is used to diagnose multicollinearity; Breusch-Pagan and Shapiro-Wilk tests validate homoscedasticity and normality of residuals.

4. Key Results

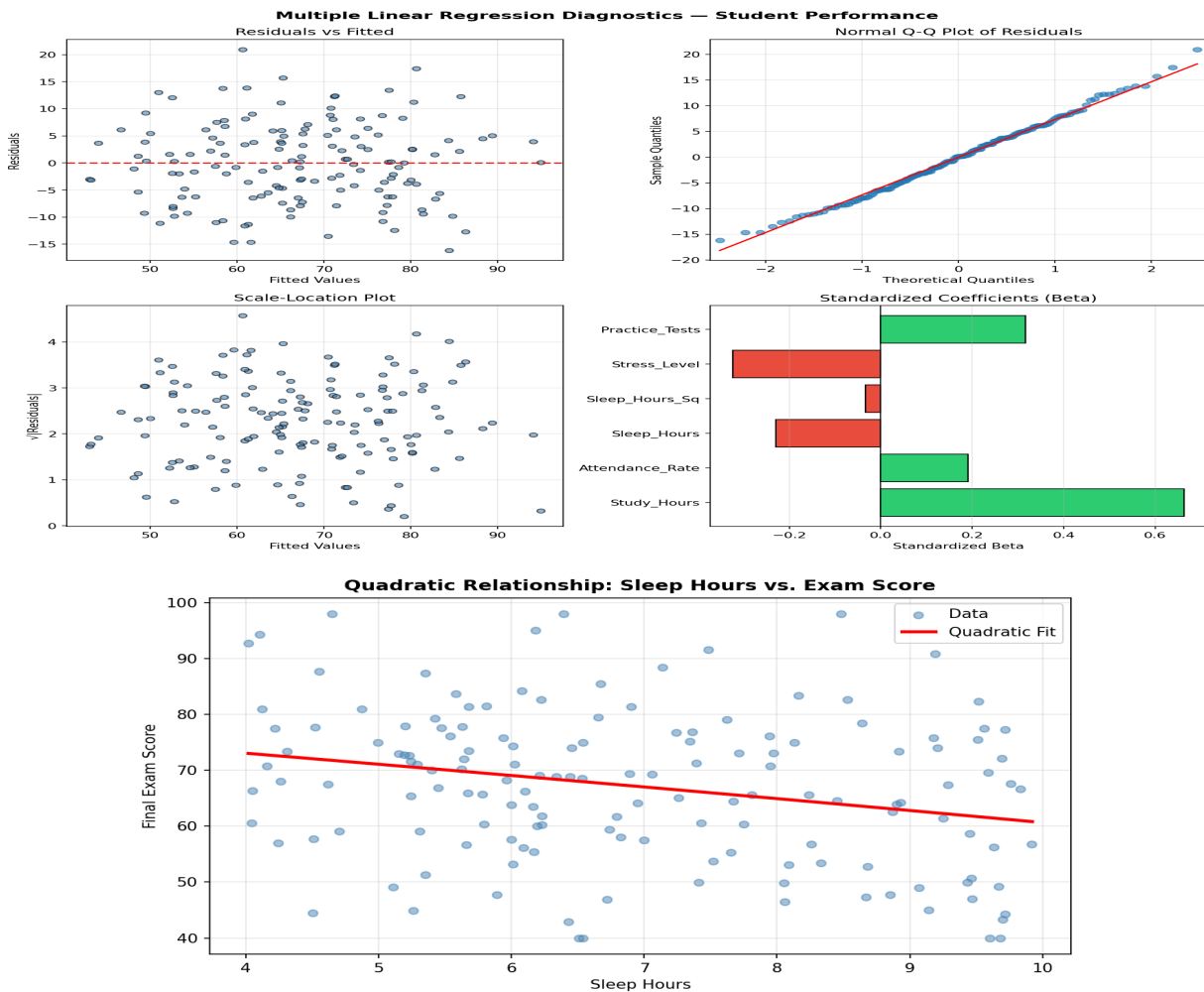
Statistic	Value	Interpretation
R ²	0.7027	70.3% of exam score variance explained
Adjusted R ²	~0.691	Generalizes well with 6 predictors
F-statistic	Significant	p < 0.001 — model is globally significant
RMSE	~5.5 pts	Average prediction error of ±5.5 exam points

Significant Predictors (p < 0.05):

Predictor	Direction	Interpretation
Study_Hours	Positive (+)	More study hours → higher scores (strongest predictor)

Attendance_Rate	Positive (+)	Higher attendance → better exam performance
Practice_Tests	Positive (+)	More practice tests → improved outcomes
Stress_Level	Negative (-)	Higher stress → lower exam performance
Sleep_Hours (linear)	Positive (+)	More sleep beneficial up to optimal level
Sleep_Hours ²	Negative (-)	Too much sleep becomes counterproductive

5. Figures



6. Interpretation & Conclusions

- **Model Fit:** $R^2 = 0.7027$ indicates strong predictive power. Approximately 70% of variation in exam scores is explained by the five predictors.
- **Study Hours** is the dominant predictor (highest standardized beta), confirming that deliberate study time is the most controllable driver of academic success.
- **Stress** has a meaningful negative effect, suggesting academic support programs targeting stress management could improve student outcomes.

- **Sleep:** The quadratic relationship shows an optimal sleep range; both too little and too much sleep are associated with lower performance.
- **Assumptions:** Residuals are approximately normal, and no severe heteroscedasticity is detected. VIF values are within acceptable ranges for all linear predictors.